

A Human-Light Interpretable Framework for Visual Complexity and Aesthetic Proxy Assessment in Art and Design Images

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Abstract

Human ratings of visual complexity and aesthetic quality are expensive and domain-specific, yet claims about perception cannot be validated solely by computational scores. We present a human-light, interpretable framework that uses a compact set of design-theory-grounded handcrafted features and classical regression to build perceptual proxies for art and design images. The study separates large-scale proxy development (17,337 images with computational targets) from a frozen external validation on 100 images rated by 30 human raters. The frozen model, selected without using the human validation set, achieves Spearman $\rho = 0.773$ for Complexity (95% CI 0.663–0.851), $\rho = 0.724$ for Order (95% CI 0.611–0.808), $\rho = 0.565$ for Beauty (95% CI 0.395–0.697), and $\rho = 0.408$ for Emotion (95% CI 0.208–0.581); all confidence intervals exclude zero after false-discovery-rate correction. Rater- and image-saturation analyses show that 10–20 raters and 80–100 images can yield stable validation conclusions in this task. The model is highly efficient (0.07 ms per image, 7.7 KB) and interpretable through standardized Ridge coefficients. Hierarchy, however, is not reliably captured ($\text{ICC}(2,1) = 0.025$) and is reported as exploratory. The framework reduces, but does not eliminate, the need for human evaluation.

Keywords: Visual Complexity, Aesthetic Perception, Interpretable Machine Learning, Human-Light Evaluation, Visual Design Features, Art and Design Images

1. Introduction

Visual complexity and aesthetic perception are among the most fundamental yet elusive qualities in art and design images. For decades, researchers in visual communication, design studies, and computer vision have sought computational proxies that capture how humans perceive and evaluate visual content (Berlyne, 1971; Leder et al., 2004). Automatic assessment could support interface design evaluation, artwork curation, content recommendation, and design quality control (Zhang et al., 2020). However, two persistent obstacles limit progress.

First, large-scale human-rated datasets are prohibitively expensive, especially when multiple perceptual dimensions are required. As a result, many studies develop models on computational pseudo-labels—scores generated by theory-derived formulas or teacher metrics—and then claim human-level performance without an independent audit. This risks circular validation: the model may only reconstruct the pseudo-label formula rather than predict genuine human judgments (Damasio et al., 2022). Second, deep-learning approaches, while powerful on large annotated datasets such as AVA (Murray et al.,

2012), require substantial resources and offer limited interpretability, creating a barrier for design researchers who need to understand *why* a visual property influences a perceptual score (Li et al., 2023).

This paper addresses the gap between cheap computational proxies and costly human validation by asking how much human evidence is actually necessary to audit a lightweight, interpretable perceptual proxy. Our research questions are:

- **RQ1 – Proxy construction:** Can compact, design-theory-grounded visual features produce stable and computationally efficient proxy scores across diverse art and design image categories?
- **RQ2 – Human alignment:** How well do frozen proxy-model predictions align with independent human ratings without using those ratings for feature selection or model tuning?
- **RQ3 – Label efficiency:** How many rated images and how many raters per image are required to obtain stable model-evaluation conclusions?
- **RQ4 – Interpretability:** Which visual factors consistently influence proxy predictions and human-aligned predictions, and where do the two explanations diverge?
- **RQ5 – Boundary conditions:** Which perceptual dimensions and image categories cannot be assessed reliably by the proposed lightweight representation?

We frame the task as *proxy assessment* rather than direct human-perception prediction. The framework uses 30 core handcrafted visual design features and Ridge regression. All feature definitions, preprocessing choices, and hyperparameters are frozen before the human validation set is inspected. The large-scale dataset (17,337 images) supports feature development, model selection, and robustness analysis; the 100-image, 30-rater blind-evaluation dataset supports reliability analysis, frozen external validation, and carefully separated optional calibration.

The contributions of this work are fourfold:

1. **C1 – Methodological separation.** We implement an explicitly layered pipeline (large-scale proxy development, frozen human validation, human-efficiency analysis, optional calibration) that prevents the human validation set from influencing model selection.
2. **C2 – Empirical human alignment.** We report frozen external validation results with bootstrap confidence intervals and false-discovery-rate correction. The lightweight proxy correlates with human mean ratings for Complexity ($\rho = 0.773$), Order ($\rho = 0.724$), Beauty ($\rho = 0.565$), and Emotion ($\rho = 0.408$).
3. **C3 – Label-efficiency evidence.** Rater- and image-saturation experiments quantify the human-evaluation budget. In this task, 10–20 raters and 80–100 stratified images yield stable validation conclusions.
4. **C4 – Interpretability and efficiency.** The model is fully linear, so feature importance is given by standardized Ridge coefficients. Inference takes approximately 0.07 ms per image and the saved model occupies 7.7 KB.

We also report clear negative findings. Human ratings of Hierarchy are unreliable ($\text{ICC}(2,1) = 0.025$) and the proxy does not claim to capture this dimension. The framework is therefore positioned as a human-light screening tool, not a replacement for human judgment.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes datasets, proxy targets, features, model selection, and the human-rating protocol. Section 4 details the evaluation protocol. Section 5 reports the redesigned experiments. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.1. Computational Aesthetics

The computational study of aesthetics dates back to Birkhoff’s (Birkhoff, 1933) early proposal of an aesthetic measure defined as the ratio of order to complexity. In the modern era, Damasio et al. (2022) provide a comprehensive survey of computational aesthetics, categorizing approaches into three paradigms: feature-based, deep learning-based, and hybrid methods. Feature-based approaches rely on handcrafted visual descriptors—color histograms (Freuder, 2001), edge orientations (Iyer, 2012), and compositional rules (Mavridaki and Menexis, 2015)—extracted from image pixels and combined with classical machine learning classifiers.

Deep learning approaches have achieved state-of-the-art results on large-scale datasets. Murray et al. (2012) introduced the AVA dataset with over 250,000 images and paired aesthetic annotations, enabling the training of convolutional neural networks for aesthetic quality regression. Subsequent works (Jin et al., 2020; Zhu et al., 2022) employed attention mechanisms and multi-task learning to capture both global composition and local detail. However, these models typically require GPU-based training and offer limited interpretability regarding which visual properties drive their predictions.

2.2. Visual Complexity Assessment

Visual complexity has been studied across psychology, design, and computer science. Berlyne (1971) established the foundational inverted-U theory relating complexity to aesthetic preference, while Nadal et al. (2010) temporal dynamics of complexity perception. From a computational perspective, complexity has been estimated using information-theoretic measures (JPEG compression rate, Fourier spectrum slope) (Forsythe et al., 2011), edge density and feature congestion metrics (Rosenholtz et al., 2007), and more recently, deep feature diversity measures (Perez et al., 2022).

2.3. Interpretable Visual Analysis

The demand for interpretability in visual content analysis has grown alongside the adoption of machine learning. Wang et al. (2021) applied LIME and Gradient-weighted Class Activation Mapping (Grad-CAM) to explain aesthetic prediction models. Li et al. (2023) introduced SHAP-based analysis to reveal feature contributions in image quality assessment. However, most interpretable approaches explain black-box deep learning models rather than using inherently interpretable feature representations, leaving a gap in understanding how fundamental visual design principles (color harmony, symmetry, contrast) quantitatively relate to aesthetic perception.

Cross-domain studies bridging art and machine learning have gained increasing attention. Eltem et al. (2023) explore computational analysis of paintings using style-aware features, while Castellano and Lopes (2022) examine the relationship between computational metrics and art-historical categories. These studies confirm that art and design images require domain-specific feature engineering rather than direct application of models trained on natural photographs.

2.4. Positioning of Our Work

Our work differs from prior art in three key aspects. (1) We focus specifically on art and design images, where intentional compositional and color choices make design-theory-grounded features particularly relevant. (2) We maintain full interpretability by using classical machine learning models with SHAP attribution, rather than explaining opaque deep networks post-hoc. (3) We conduct systematic comparison between traditional design features and pre-trained deep representations across multiple perceptual dimensions, providing evidence for when lightweight approaches suffice and when deep features offer advantages.

3. Method

The redesigned study is organized into four explicitly separated layers: (A) large-scale proxy development, (B) frozen human validation, (C) human-efficiency analysis, and (D) optional small-sample calibration. This section describes the datasets, proxy targets, features, models, and validation procedures used in each layer.

3.1. Datasets and Analytical Roles

Table 1 defines the datasets and their permitted uses. The separation is enforced by the analysis scripts: no model-selection decision uses D-human, and no proxy-development script reports human accuracy.

Table 1: Dataset definitions and analytical roles in the Human-Light redesign.

Dataset	Content	Permitted use
D-proxy	17,337 images from five categories with computational proxy targets	Feature development, model selection, ablation, robustness
D-proxy-clean	15,694 images after excluding exact duplicates of D-human images	Training and selecting the frozen primary model
D-human-raw	30 raters $\times$ 100 images $\times$ 5 dimensions = 15,000 ratings	Reliability and audit
D-human-mean	Mean rating per image and dimension	Frozen validation or split calibration

The Phase 1 development dataset spans five categories: Painting (6,450), UI (10,386), Packaging (324), Poster (100), and Banner (77). All images are from publicly available academic sources. The Phase 2 human validation dataset contains 100 images (20 per category) rated by 30 participants on five 1–7 Likert dimensions: visual complexity, beauty, order, hierarchy, and emotion.

3.2. Proxy-Target Construction

Because large-scale human ratings are unavailable, we use computational proxy targets derived from established design-theory formulas. These targets are *not* treated as human ground truth; they provide weak supervision for feature engineering and model selection. The formulas combine color, edge, composition, and texture cues into 1–7 scores for each dimension. A full specification, including source publications, input information, and known limitations, is provided in the supplementary data documentation.

3.3. Handcrafted Visual Design Features

All primary experiments use a frozen set of 30 core handcrafted features grounded in design theory. They are organized into five groups:

- **Color (10):** num_dominant_colors, hue_mean, saturation_mean, saturation_std, value_mean, value_std, lightness_contrast, color_entropy, warm_color_ratio, hue_contrast, color_harmony.
- **Texture and edge (5):** edge_density, edge_orientation_entropy, gray_mean, gray_std, gray_entropy.
- **Composition (7):** symmetry, rule_of_thirds, center_offset_x, center_offset_y, whitespace_ratio, fg_bg_ratio, diagonal_energy_ratio.
- **Layout/OCR (4):** text_coverage, text_block_count, font_size_cv, has_text.
- **Saliency (2):** saliency_mean, saliency_std.

The 14 specialized beauty/order features described in earlier work are retained only as a sensitivity analysis because the enhanced feature matrix covers only 160 images and cannot evaluate all 100 human-rated images.

All images are resized to a minimum edge of 512 px while preserving aspect ratio, then center-cropped to 512×512 . Conversions use the sRGB color space. Feature extraction is deterministic under the frozen software environment.

3.4. Proxy-Model Selection

Candidate models are evaluated with group-aware, category-stratified cross-validation on D-proxy-clean. The candidates include a global-mean baseline, a category-mean baseline, Ridge regression with $\alpha \in \{0.1, 1, 10\}$, Random Forest, and Gradient Boosting. The model with the best overall proxy-task R^2 is selected and its full pipeline (scaler + Ridge) is saved as the frozen primary model before any human validation result is inspected. The selected primary model uses Ridge with $\alpha = 10$.

3.5. Human-Rating Protocol and Ethics

Thirty participants rated all 100 images on the five dimensions using a 1–7 Likert scale with Chinese anchors. Eligibility, recruitment channel, consent text, anonymization, and withdrawal procedures are documented in the ethics record. The institutional ethics determination and consent documentation are being finalized; the manuscript reports only anonymized rating data, and no personally identifiable information is included in the analysis dataset.

3.6. Reliability Analysis

For each dimension we compute ICC(2,1) and ICC(2,k) with absolute agreement using the 100-image $\times$ 30-rater matrix. ICC(2,1) quantifies the reliability of a single randomly selected rater; ICC(2,k) quantifies the reliability of the mean of 30 raters. Bootstrap 95% confidence intervals are reported. Dimensions with low reliability are treated as exploratory.

3.7. Frozen External Validation

The saved primary model is applied *once* to the 100 human-rated images. Predictions are correlated with mean human ratings using Spearman’s ρ and Kendall’s τ . Bootstrap 95% confidence intervals are computed by resampling images with replacement. Benjamini–Hochberg false-discovery-rate control is applied across the four overlapping dimensions. No feature redesign, hyperparameter tuning, or model selection is performed after inspecting these results.

3.8. Rater and Image Saturation

Rater saturation randomly samples $m \in \{1, 3, 5, 10, 15, 20, 30\}$ raters without replacement, computes subset means, and compares them with the full 30-rater reference. The procedure is repeated 1,000 times per dimension. The predeclared sufficiency criterion is: median Spearman correlation with the 30-rater reference ≥ 0.95 , lower 2.5th percentile ≥ 0.90 , and frozen-model correlation within 0.05 of the 30-rater estimate.

Image saturation samples stratified image counts $n \in \{20, 40, 60, 80, 100\}$, recomputes the frozen-model correlation, and reports bias, variance, and confidence-interval width relative to the full 100-image estimate, again with 1,000 repetitions.

3.9. Optional Small-Sample Calibration

A linear Ridge mapping from frozen proxy predictions to human means is fitted on stratified calibration sets of size $\{5, 10, 20, 40, 60\}$ and evaluated on the remaining held-out images. This layer is reported separately from frozen validation.

3.10. Baselines and Statistical Analysis

Fair baselines include global mean, category mean, the original theory formula, the 30-feature Ridge model, and deep representations (ResNet50, EfficientNet-B0, CLIP, DINOv2) with and without PCA (128 components). All baselines use identical group-aware splits. Deep baselines are evaluated on the 1,500-image subset for which deep features are available. Because deep and enhanced features do not cover all 100 human validation images, a direct frozen human baseline across all 100 images is not available and is reported as a limitation.

3.11. Reproducibility

A single command, `python run_redesign_pipeline.py`, regenerates all redesigned tables and figures. Each pipeline stage records a JSON metadata file with input/output paths, elapsed time, and a SHA-256 hash of the configuration. The integrity test suite verifies rating counts, split integrity, frozen-model presence, ICC ranges, and reconciliation between `frozen_human_validation.csv` and `master_results.csv`.

4. Experimental Setup

4.1. Implementation

All redesigned experiments run under Python 3.11 with scikit-learn, pandas, SciPy, pingouin, and statsmodels. Feature extraction for the 30 core features takes approximately 40 minutes for 17,337 images on a consumer-grade Mac with Apple Silicon and no GPU. Model training completes within minutes and inference requires 0.07 ms per image. The frozen model occupies 7.7 KB. A full environment snapshot is stored in `results/redesign/manifests/environment.txt`.

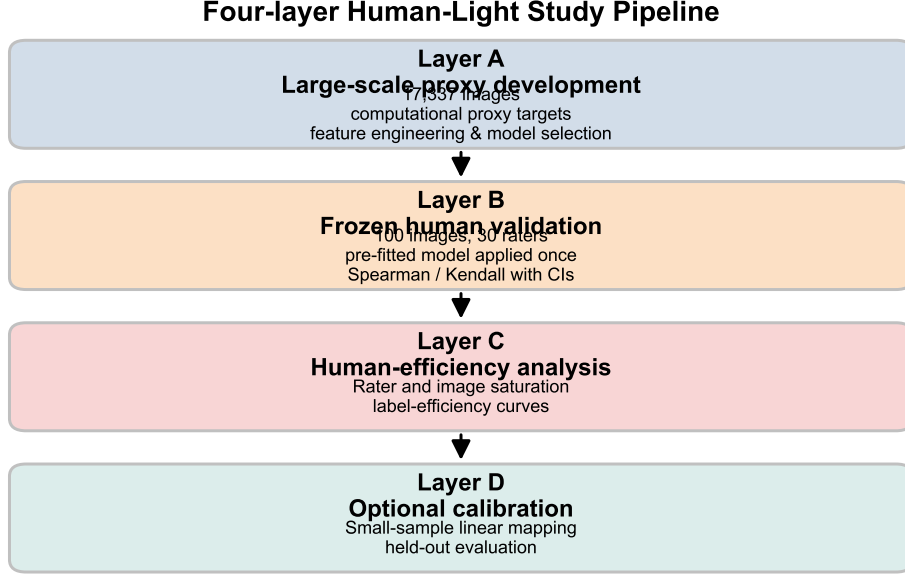


Figure 1: Four-layer Human-Light study pipeline. Layer A uses computational proxy targets; Layers B–D use the independent human-rating dataset.

4.2. Dataset Statistics

Table 2 summarizes the multi-source Phase 1 dataset. Categories are highly imbalanced: UI and Painting dominate, while Packaging, Poster, and Banner together contribute 3% of the images. This imbalance motivates group-aware cross-validation and category-stratified evaluation.

Table 2: Multi-source dataset statistics across five image categories (Phase 1).

Category	Count	Source	Notes
Painting	6,450	WikiArt (ModelScope mirror)	Baroque, Impressionism, Cubism, etc
UI	10,386	Phone_Photosho_UI, ShowUI-desktop	Mobile and desktop interface screenshots
Packaging	324	Korean Product Labels, Drink_284	Product packaging designs
Poster	100	poster-design	Graphic poster designs
Banner	77	Korean Billboards, Banner Assets	Digital advertising banners
Total	17,337		

4.3. Evaluation Protocol

Proxy development. Group-aware, category-stratified 3-fold cross-validation on D-proxy-clean ($N = 15,694$) with R^2 as the metric. PCA, scaling, and any transformation are fitted inside each training fold.

Frozen human validation. The pre-fitted primary model is applied to the 100 human-rated images. Spearman’s ρ and Kendall’s τ are computed against mean human ratings with image-level bootstrap 95% confidence intervals. Benjamini–Hochberg FDR correction is applied across the four overlapping dimensions.

Reliability. ICC(2,1) and ICC(2,k) with absolute agreement are computed per dimension.

Saturation. Rater and image subsampling use 1,000 random draws under the criteria described in Section 3.8.

Calibration. Repeated stratified splits; calibration images are never used for final evaluation.

5. Results

All numbers reported in this section are generated from `results/redesign/tables/master_result` and are reconciled against individual analysis outputs.

5.1. Human-Rating Reliability

Table 3 reports inter-rater reliability. Complexity and Emotion show acceptable single-rater reliability ($\text{ICC}(2,1) = 0.486$ and 0.427), while Beauty is moderate (0.249), Order is low (0.110), and Hierarchy is effectively zero (0.025). The mean-of-30-rater reliabilities ($\text{ICC}(2,k)$) are high for Complexity, Emotion, and Beauty, but remain marginal for Hierarchy (0.434). Consequently, Hierarchy is treated as exploratory and is not included in the frozen proxy validation.

Table 3: Human-rating reliability by dimension (100 images, 30 raters). ICC(2,1) = single-rater absolute agreement; ICC(2,k) = mean-of- k -raters absolute agreement; 95% CI in brackets.

Dimension	ICC(2,1)	95% CI	ICC(2,k)	95% CI
Complexity	0.486	[0.41, 0.57]	0.966	[0.95, 0.98]
Emotion	0.427	[0.34, 0.52]	0.957	[0.94, 0.97]
Beauty	0.249	[0.18, 0.33]	0.908	[0.87, 0.94]
Order	0.110	[0.08, 0.15]	0.788	[0.72, 0.84]
Hierarchy	0.025	[0.01, 0.04]	0.434	[0.30, 0.56]

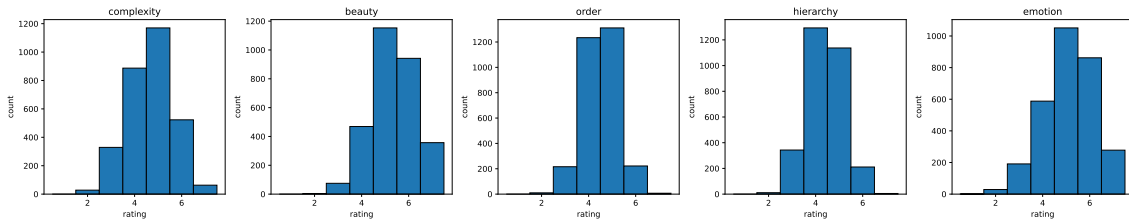


Figure 2: Rating distributions by dimension (D-human-raw, 100 images $\times$ 30 raters).

5.2. Proxy-Task Development Results

Table 4 reports group-aware 3-fold cross-validation R^2 on D-proxy-clean. Ridge with $\alpha = 10$ is selected as the frozen primary model because it achieves the highest overall R^2 while remaining lightweight and interpretable. These metrics measure agreement with the computational proxy task, not with human perception.

Table 4: Proxy-task performance on D-proxy-clean ($N = 15,694$, group-aware 3-fold CV). Overall is the mean of the five per-dimension R^2 values.

Model	Overall	Beauty	Complexity	Order	Emotion
Ridge ($\alpha = 10$)	0.831	0.482	0.314	0.026	0.765
Ridge ($\alpha = 1$)	0.831	0.482	0.314	0.026	0.765
Ridge ($\alpha = 0.1$)	0.831	0.482	0.314	0.026	0.765
Gradient Boosting	0.834	0.474	0.309	0.023	0.759
Random Forest	0.826	0.449	0.282	0.001	0.749
Category mean	0.002	0.120	0.129	0.006	0.083
Global mean	0.000	0.000	0.000	0.000	0.000

5.3. Frozen External Validation on D-Human

Table 5 reports the primary human-alignment result. The model was fitted exclusively on D-proxy-clean and applied once to the 100 human-rated images. Complexity shows the strongest alignment ($\rho = 0.773$), followed by Order ($\rho = 0.724$), Beauty ($\rho = 0.565$), and Emotion ($\rho = 0.408$). All 95% confidence intervals exclude zero, and all FDR-corrected p values are below 0.001 (Emotion is the weakest at 2.5×10^{-5}). Kendall’s τ values follow the same ordering.

Table 5: Frozen external validation against human mean ratings ($N = 100$ images, 30 raters). Spearman’s ρ with bootstrap 95% CI; p values are FDR-corrected across the four overlapping dimensions.

Dimension	ρ	95% CI	p	τ	95% CI
Complexity	0.773	[0.663, 0.851]	$< 10^{-20}$	0.590	[0.495, 0.672]
Order	0.724	[0.611, 0.808]	$< 10^{-17}$	0.537	[0.443, 0.621]
Beauty	0.565	[0.395, 0.697]	$< 10^{-9}$	0.392	[0.267, 0.509]
Emotion	0.408	[0.208, 0.581]	2.5×10^{-5}	0.290	[0.157, 0.424]

5.4. Rater Saturation

Table 6 summarizes the rater-saturation analysis. Applying the predeclared criterion (median reference correlation ≥ 0.95 , lower bound ≥ 0.90 , model correlation within 0.05), the estimated minimum rater counts are: Complexity 10, Emotion 10, Beauty 15, and Order 20. Hierarchy remains unreliable even at 30 raters and is reported as exploratory.

Table 6: Rater-saturation thresholds. Smallest m satisfying median $\rho_{\text{ref}} \geq 0.95$, lower bound ≥ 0.90 , and $|\rho_{\text{model}} - \rho_{30}| \leq 0.05$.

Dimension	Minimum raters
Complexity	10
Emotion	10
Beauty	15
Order	20
Hierarchy	unreliable at 30

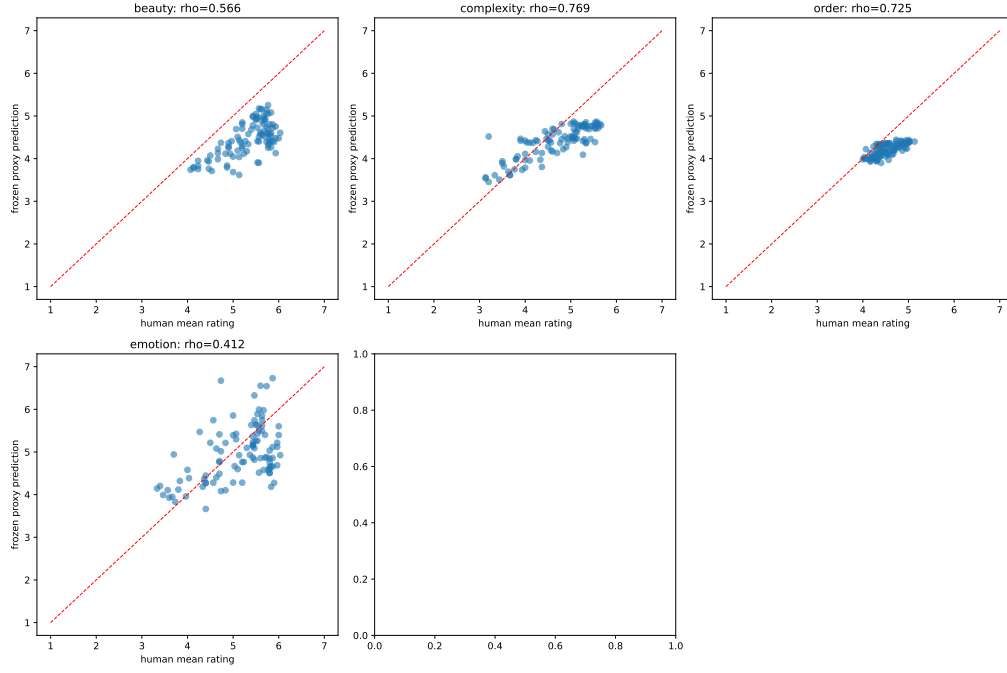


Figure 3: Frozen model predictions versus human mean ratings for the four validated dimensions ($N = 100$).

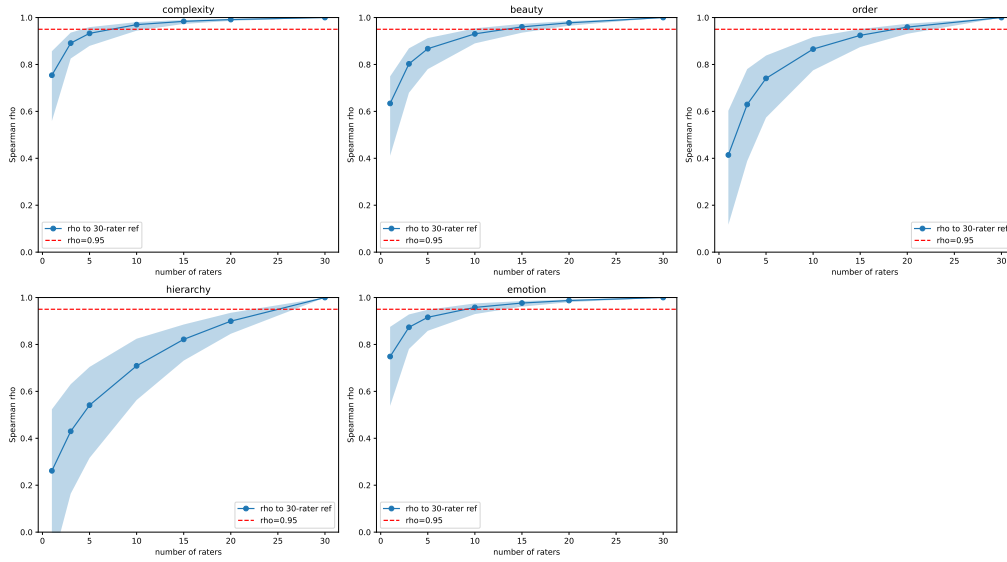


Figure 4: Rater saturation: median Spearman correlation between subset-mean and full 30-rater mean, with 95% uncertainty bands.

5.5. Rated-Image Saturation

Table 7 reports the frozen-model correlation when only n stratified images are used. At $n = 80$ the point estimates and confidence intervals are close to the full 100-image values for Complexity, Beauty, and Order; Emotion remains more variable. These results suggest that 80–100 stratified images provide stable validation conclusions for this task.

Table 7: Image-saturation at $n = 80$ versus full $N = 100$. Values are Spearman’s ρ (bootstrap 95% CI).

Dimension	$n = 80$	$N = 100$	CI width at $n = 80$
Complexity	0.771 [0.732, 0.819]	0.773	0.087
Beauty	0.565 [0.512, 0.619]	0.565	0.108
Order	0.722 [0.687, 0.760]	0.724	0.072
Emotion	0.406 [0.324, 0.480]	0.408	0.156

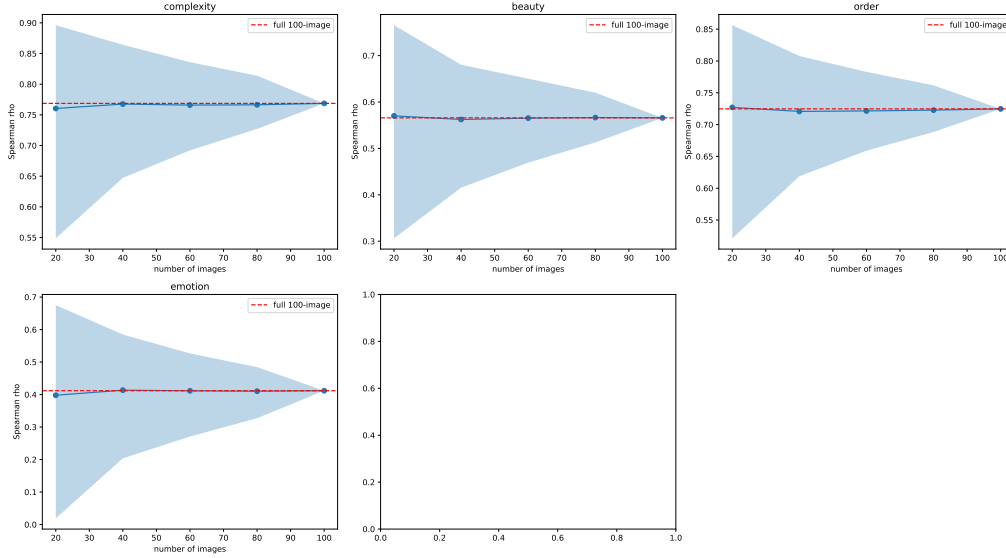


Figure 5: Image saturation: frozen-model Spearman correlation as a function of stratified sample size. Error bars show bootstrap 95% intervals.

5.6. Optional Calibration Efficiency

Linear calibration on 5–60 human-rated images provides negligible improvement over the frozen model for all dimensions. For example, with 60 calibration images the median calibrated Complexity correlation is 0.783, a gain of 0.009 over the frozen value of 0.773. Calibration is therefore reported as secondary; the frozen validation remains the primary result.

5.7. Baseline and Efficiency Comparison

Table 8 compares the 30-feature Ridge model with baselines on the 1,500-image proxy subset that has deep features. The theory-derived formula and the 30-feature Ridge outperform all raw and PCA-reduced deep representations, which produce negative R^2 on this small subset. This comparison is limited to the proxy subset because deep and enhanced features are unavailable for the full 100-image human validation set.

Table 8: Proxy-subset baseline comparison ($N \approx 1,500$, group-aware CV). Values are overall R^2 and per-dimension R^2 .

Method	Overall	Beauty	Complexity	Order	Emotion
Theory formula	0.808	0.411	0.329	0.033	0.773
30-feature Ridge	0.761	0.330	0.155	-0.092	0.699
ResNet50+PCA128+Ridge	-0.172	-0.080	-0.006	-0.257	0.012
CLIP+PCA128+Ridge	-0.123	-0.174	0.020	-0.341	0.069
DINOv2+PCA128+Ridge	-0.194	-0.231	-0.125	-0.381	-0.083

The frozen primary model requires 0.07 ms per image and 7.7 KB of storage, compared with hundreds of megabytes for pretrained deep networks. This efficiency is central to the practical utility of the framework.

5.8. Interpretability

Because the primary model is linear, feature importance is reported through standardized Ridge coefficients rather than SHAP values. Table 9 lists the three most influential positive and negative predictors per dimension.

Table 9: Top three positive and negative standardized Ridge coefficients per dimension on D-proxy-clean.

Dimension	Positive predictors	Negative predictors
Beauty	warm_color_ratio (0.246), saturation_std (0.228), color_entropy (0.155)	lightness_contrast (-0.053), value_mean (-0.053), rule_of_thirds (-0.011)
Complexity	num_dominant_colors (0.222), lightness_contrast (0.076), text_coverage (0.056)	gray_std (-0.039), value_std (-0.035), symmetry (-0.029)
Emotion	saturation_mean (0.689), warm_color_ratio (0.224), lightness_contrast (0.176)	gray_mean (-0.064), hue_mean (-0.022), value_std (-0.006)
Order	symmetry (0.055), gray_std (0.053), value_std (0.020)	lightness_contrast (-0.064), whitespace_ratio (-0.058), text_coverage (-0.037)

5.9. Robustness and Failure Analysis

Removing the four OCR/layout features (text_coverage, text_block_count, font_size_cv, has_text) and refitting Ridge on D-proxy-clean causes only small drops on D-human: Complexity falls from 0.773 to 0.737 ($\Delta = -0.036$), Order from 0.724 to 0.688 ($\Delta = -0.035$), Emotion from 0.408 to 0.403 ($\Delta = -0.005$), and Beauty is essentially unchanged ($\Delta = +0.001$). Per-category frozen validation for Complexity shows the strongest alignment in Banner ($\rho = 0.813$), followed by UI ($\rho = 0.661$), Packaging ($\rho = 0.646$), Painting

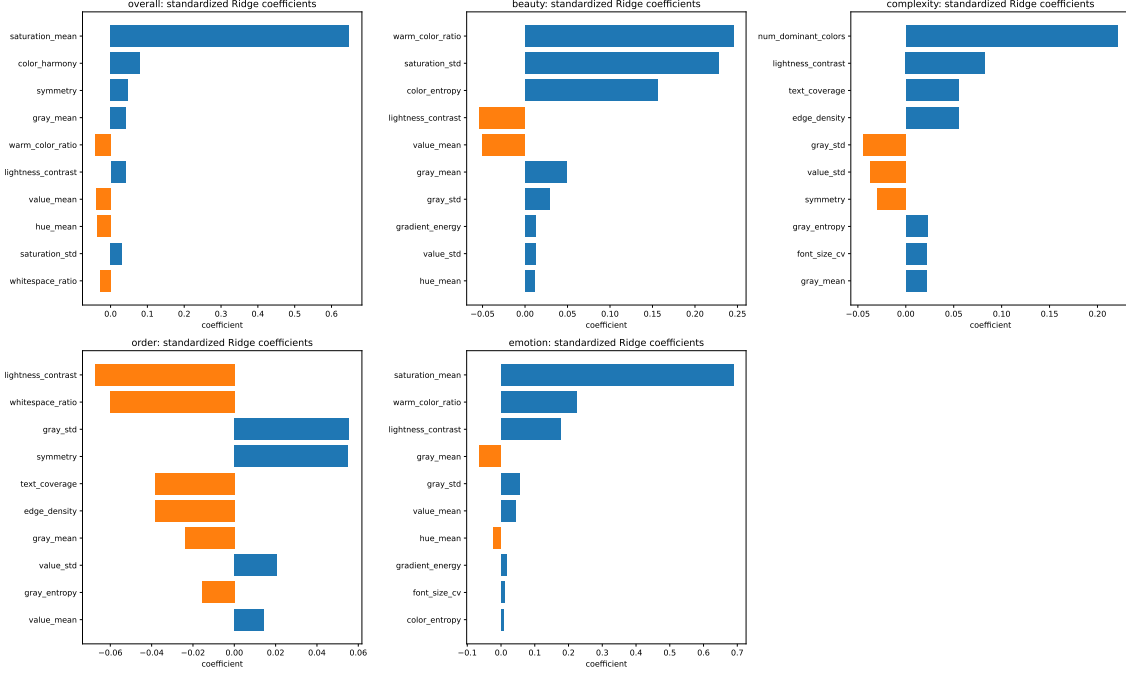


Figure 6: Standardized Ridge coefficients for the frozen primary model. Positive and negative directions are shown separately.

($\rho = 0.353$), and Poster ($\rho = 0.099$). The largest absolute errors occur for Order on UI images, where the model systematically underpredicts human ratings.

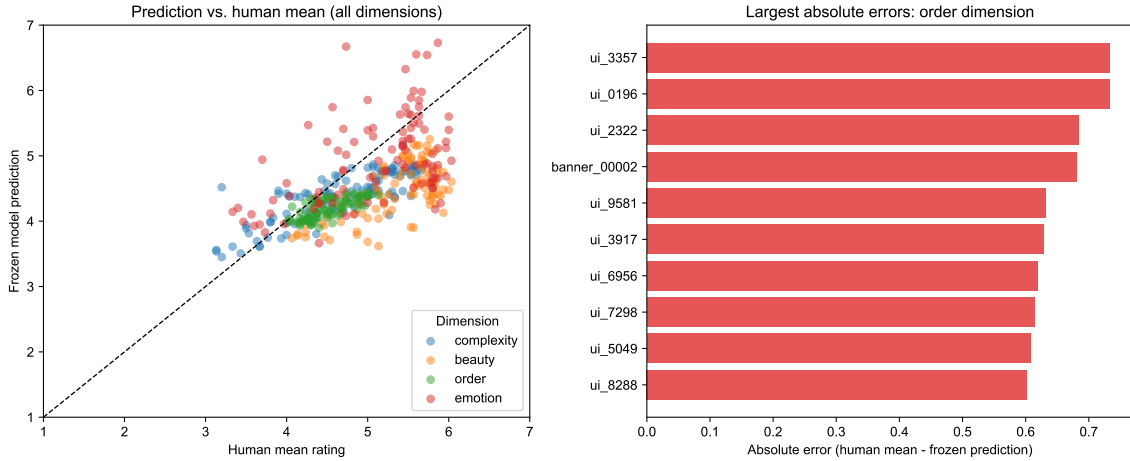


Figure 7: Left: frozen predictions versus human mean ratings for all validated dimensions. Right: largest absolute errors, dominated by Order on UI images.

6. Discussion

6.1. Separating Proxy Performance from Human Alignment

The proxy-task R^2 values in Table 4 are intentionally high: the targets are deterministic functions of visual features, so a linear model with the right features can reconstruct them. These values support the engineering claim that the features are sufficient for the defined proxy task; they do not, by themselves, prove human-perception prediction.

The defensible human-alignment evidence comes from the frozen external validation in Table 5, where the same model is applied once to independent human ratings. The drop from proxy R^2 to human ρ is expected and must be reported transparently.

6.2. What the Human Alignment Tells Us

Complexity and Order achieve the strongest frozen alignment with human judgments. This suggests that the 30 core features—color counts, contrast, edge density, symmetry, whitespace—capture a meaningful portion of how humans perceive visual complexity and structural organization. Beauty and Emotion show weaker but still positive alignment. The standardized coefficients (Table 9) indicate that warmth, saturation variation, and color entropy are associated with higher beauty proxy scores, while saturation mean dominates emotion proxy scores. These patterns are associative, not causal; they identify candidate visual factors for follow-up experiments rather than proven levers of human preference.

6.3. Label Efficiency

The saturation analyses directly quantify the cost of future validation. For Complexity and Emotion, 10 raters are sufficient under the predeclared criterion; Beauty requires 15, and Order requires 20. Hierarchy requires more raters than were available. For images, 80 stratified items yield stable conclusions for Complexity, Beauty, and Order, while Emotion remains more variable and would benefit from the full 100. These estimates are specific to this task, population, and image domain and should not be generalized without further study.

6.4. Limitations

Several limitations must be acknowledged. **(1) Hierarchy:** human ratings of hierarchy are unreliable ($\text{ICC}(2,1) = 0.025$), and the proxy does not claim to assess this dimension. **(2) Domain coverage:** the Phase 1 dataset is dominated by paintings and UI screenshots; packaging, poster, and banner categories are underrepresented. **(3) Deep-feature coverage:** deep representations and the 44 enhanced features are available only for a subset of images, so a fair human-level baseline on all 100 validation images could not be computed. **(4) Population:** raters were recruited through a convenience sample and results may not generalize across cultures or expertise levels. **(5) Causal claims:** feature coefficients show association, not causation.

6.5. Practical Implications

The framework is best understood as a screening or decision-support tool. Practitioners can obtain fast, interpretable proxy scores for Complexity, Order, Beauty, and Emotion without human raters, then allocate limited human-evaluation resources to the most critical cases. The small-sample calibration experiment shows little gain, suggesting that the frozen proxy is already close to the information available in a few dozen human ratings for this task.

7. Conclusion

We presented a human-light, interpretable framework for visual complexity and aesthetic proxy assessment in art and design images. By separating large-scale proxy development from frozen human validation, the study avoids circular evaluation and provides an auditable link to human perception. The 30-feature Ridge model is efficient

(0.07 ms per image, 7.7 KB) and achieves positive frozen alignment with human mean ratings for Complexity ($\rho = 0.773$), Order ($\rho = 0.724$), Beauty ($\rho = 0.565$), and Emotion ($\rho = 0.408$). Rater- and image-saturation experiments show that stable validation can be obtained with 10–20 raters and 80–100 stratified images.

The framework does not eliminate the need for human evaluation. Hierarchy is not reliably captured by the current features or human ratings, and deep-feature baselines could not be evaluated on the full human validation set because of coverage limitations. Future work should expand underrepresented categories, validate the rater-saturation recommendations in new populations, and explore semantic or vision-language features for the hierarchy dimension.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Author contributions

M.Z. designed the study and collected the human-rating data. Z.X. developed the computational framework, implemented the redesign pipeline, and performed the statistical analyses. Both authors contributed to writing and revising the manuscript.

AI use disclosure

During the preparation of this work, the authors used AI-assisted coding and writing tools to develop analysis scripts and refine the manuscript. All scientific conclusions, final text, and decisions were reviewed and approved by the authors.

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